

To the attention of:
INFN – CSN5
Young Researchers Grant Selection Committee

Subject: Letter of support for Dr Simone Manti's PRISM project application to the INFN–CSN5 Young Researchers Grant programme

Due2Lab S.r.l. is pleased to express its support for PRISM ("Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements"), a project led by Dr Simone Manti of INFN – Laboratori Nazionali di Frascati.

Due2Lab designs and manufactures CdZnTe (CZT) sensors and detector systems for X-ray and gamma-ray applications, including single-pixel devices, linear arrays and pixel matrices. From this perspective, we recognize a strong technological connection with PRISM, which aims to establish a laboratory fluorescence-XAS platform above 20 keV based on optimized crystal optics and a custom multichannel CZT detector operating at room temperature.

PRISM addresses detector requirements that are especially relevant to the development of advanced CZT instrumentation: high hard-X-ray efficiency, stable spectroscopic response, channel-to-channel uniformity, operation at useful counting rates, and control of charge-collection, dead-time and pile-up effects. The project's plan to characterize the complete detection chain, from the sensor and low-noise front-end electronics to calibration and waveform-based spectral reconstruction, can provide valuable knowledge for future scientific and industrial applications of segmented CZT systems.

We also consider the proposed integration of the detector with a compact laboratory spectrometer to be of particular interest. Demonstrating quantitative fluorescence-XAS on reference and realistic samples would extend the application of room-temperature CZT technology beyond stand-alone radiation detection and towards complete analytical instrumentation for materials characterization.

Due2Lab is pleased to support the scientific and technological objectives of PRISM and is interested in following its progress and exploring possible future technical exchanges concerning CZT sensor architectures, detector characterization and application requirements. Any specific contribution or collaboration would be discussed separately and subject to mutual agreement.

We believe that PRISM can strengthen the link between semiconductor-detector development and practical hard-X-ray spectroscopy, creating valuable opportunities for research, innovation and technology transfer.

Please do not hesitate to contact us for any further information.

Yours sincerely,

Scandiano, 28/07/2026



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